

NextTopDocker: the largest-to-date docking power benchmark reveals that deep learning performs generally much worse than logistic regression models

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Abstract

Predicting three-dimensional binding orientations of drug-like molecules remains challenging in structure-based drug design. Following AlphaFold's success, end-to-end machine-learning (ML) docking frameworks have emerged, leveraging deep learning for pose generation. Yet, their performance is often assessed on small, outdated benchmarks containing only a few hundred protein-ligand complexes. We present "NextTopDocker," a large, up-to-date, openaccess data set comprising 14,038 training and 5,201 test entries across 3,173 unique protein targets, defined by a cold-ligand split and prepared from the Protein Data Bank. Developed entirely with open-source tools, it includes crystallographic structures, Smina-generated docking poses, and ligand-similarity-aware training subsets. With "NextTopDocker," we benchmarked four state-of-the-art end-to-end ML docking frameworks, DeepDock, Interformer, SurfDock, and Uni-Mol Docking v.2, against classical (Smina) and hybrid (GNINA 1.3 and logistic regression using Smina and GNINA 1.3 scores) baselines. Despite sophisticated architectures, only Interformer matched the docking power of logistic regression on Smina poses, while others failed without post-docking physics-based refinement. Most raw MLgenerated poses exhibited steric clashes and/or implausible geometries, underscoring the lack of intrinsic physical realism in current ML-based docking. Embedding physics-informed constraints directly into pose generation, rather than relying on downstream optimization, is essential for achieving valid, autonomous molecular docking. "NextTopDocker" is available at <https://github.com/caominhtr/NextTopDocker> and <https://zenodo.org/records/17492994>.

Keywords

molecular docking, docking power, pose sampling, machine-learning, pose plausibility, PoseBusters, logistic regression

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Abstract

Predicting three-dimensional binding orientations of drug-like molecules remains challenging in structure-based drug design. Following AlphaFold's success, end-to-end machine-learning (ML) docking frameworks have emerged, leveraging deep learning for pose generation. Yet, their performance is often assessed on small, outdated benchmarks containing only a few hundred protein-ligand complexes. We present "NextTopDocker," a large, up-to-date, open-access data set comprising 14,038 training and 5,201 test entries across 3,173 unique protein targets, defined by a cold-ligand split and prepared from the Protein Data Bank. Developed entirely with open-source tools, it includes crystallographic structures, Smina-generated docking poses, and ligand-similarity-aware training subsets. With "NextTopDocker," we benchmarked four state-of-the-art end-to-end ML docking frameworks, DeepDock, Interformer, SurfDock, and Uni-Mol Docking v.2, against classical (Smina) and hybrid (GNINA 1.3 and logistic regression using Smina and GNINA 1.3 scores) baselines. Despite sophisticated architectures, only Interformer matched the docking power of logistic regression on Smina poses, while others failed without post-docking physics-based refinement. Most raw ML-generated poses exhibited steric clashes and/or implausible geometries, underscoring the lack of intrinsic physical realism in current ML-based docking. Embedding physics-informed constraints directly into pose generation, rather than relying on downstream optimization, is essential for achieving valid, autonomous molecular docking. "NextTopDocker" is available at <https://github.com/caominhtr/NextTopDocker> and <https://zenodo.org/records/17492994>.

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1. Introduction

Determining the three-dimensional (3D) binding orientation of a drug-like molecule and the molecular interactions stabilizing its complex with a receptor remains a central challenge in structure-based (SB) drug discovery.¹⁻⁴ It represents the first and most critical stage of virtual screening (VS) workflows aimed at identifying bioactive candidates. Over the past decades, numerous computational strategies, most notably molecular docking, have been developed to predict plausible ligand binding conformations.^{5,6} These approaches rely on pose sampling algorithms that explore ligand conformational and orientational space, with or without receptor flexibility, thus enabling rigid-body or flexible docking schemes. Sampling strategies range from shape matching and incremental construction to stochastic searches such as Monte Carlo or simulated annealing.⁵ Historically, such methods have been able to recover at least one near-native binding pose, defined as a conformation within 2-Å root-mean-square deviation (RMSD) of the experimentally determined structure.² Following pose generation, pose scoring is performed to evaluate each ligand-receptor complex. Classical scoring functions (SFs), whether physics-based, knowledge-based, or empirical, serve this purpose but frequently fail to correctly prioritize native-like docking poses, as shown in retrospective evaluations.⁷⁻¹⁰ To address these limitations, machine-learning (ML)-based SFs have emerged, leveraging data-driven approaches to rescore docking output. Examples include RF-Score-VS,¹¹ the GNINA convolutional neural network (CNN)-based SF and its successors,^{12,13} as well as the recently developed ClassyPose,¹⁰ all of which have demonstrated superior docking power compared with traditional, fully non-ML docking protocols that combine conventional pose sampling and pose scoring.

Since the introduction of AlphaFold,¹⁴ a new generation of end-to-end ML docking tools has emerged, integrating ML for both pose sampling and scoring. Among the early entrants, DeepDock (2021) applies geometric deep learning (DL) to predict ligand binding conformations: it learns a tailored statistical potential based on distance likelihoods for each ligand-target pair, thereby combining statistical modeling with SB reasoning.¹⁵ In the subsequent years, especially following AlphaFold 3 and its related innovations (e.g., Boltz-1, Boltz-2),¹⁶⁻¹⁸ several more end-to-end ML docking models have appeared. In November 2024, Interformer was released: it uses a graph-transformer backbone to model non-covalent interactions via an interaction-aware mixture density network and incorporates a negative sampling strategy to improve the predicted distribution of interactions across ligand-protein pairs.¹⁹ In 2025, SurfDock and Uni-Mol Docking v.2 further extended the field.^{20,21} SurfDock

integrates protein sequence, 3D structure and surface features within an equivariant architecture, and uses a diffusion model on a non-Euclidean manifold to optimize translation, rotation and torsion for reliable binding pose generation.²⁰ Uni-Mol Docking v.2 is an upgraded version of Uni-Mol Docking, a universal molecular representation learning framework.²¹ It uses a transformer architecture equivariant to 3D rotations and translations, shared by its molecular and pocket pre-training branches, enhancing practical representation capacity.²¹

However, the literature has raised concerns about the chemical and physical plausibility of poses generated by such ML docking methods.^{22–24} The seminal PoseBusters study, in particular, argued that artificial intelligence (AI)-based docking methods often fail to produce physically valid conformations or generalize well to novel proteins.²² Moreover, the docking power of these tools has typically been benchmarked on small and somewhat outdated data sets such as the Astex Diverse set (85 entries),²⁵ CASF-2016 (285 entries),²⁶ PDBbind core sets (a few hundred entries),²⁷ and PoseBusters (308 entries).²² Larger and up-to-date benchmarks like PDBbind are not fully open-access: as of August 2025, the latest freely available version is PDBbind v.2020R1, which was last updated in 2020 and lacks recent Protein Data Bank (PDB) entries. Furthermore, many PDBbind entries were processed using proprietary software (e.g., Sybyl),²⁷ which limits reproducibility and free reprocessing by the community.

In light of these limitations, our study has four main objectives:

- (i) To construct a large-scale, up-to-date, and meticulously curated open-access benchmarking data set of protein-ligand complexes retrieved from the PDB as of May 2025, using exclusively open-source tools. The data set was partitioned into a training set (over 14,000 entries) and a test set (over 5,000 entries) via a strict cold-ligand split (no duplicating ligands across the two partitions), and further divided into nine training subsets following a ligand-similarity-aware scheme, enabling both performance evaluation and future training of ML-based docking/scoring tools;
- (ii) To benchmark the docking power of four state-of-the-art (SOTA) end-to-end ML docking tools, DeepDock, Interformer, SurfDock, and Uni-Mol Docking v.2, on our large test set, representing the largest-scale evaluation of ML-based docking power to date and the first joint assessment of these tools on such extensive test data;
- (iii) To compare these ML methods against more traditional or hybrid approaches: pure non-ML docking (Smina),²⁸ CNN-based rescoring of Smina poses (GNINA 1.3),¹³ and logistic regression models trained on Smina and GNINA 1.3 scores across our training partitions. Docking performance was assessed using the RMSD of the top-ranked pose per PDB entry, together with PoseBusters-based physicochemical plausibility evaluation,²²

(iv) To reveal the limitations of current SOTA end-to-end ML docking methods, particularly in their intrinsic pose sampling capability, by explicitly excluding any post-generation physics-guided or non-ML optimization (e.g., energy minimization), ensuring that performance reflects the true generative power of their ML sampling algorithms.

All data and code are available on GitHub (<https://github.com/caominhtr/NextTopDocker>) and Zenodo (<https://zenodo.org/records/17492994>). On another note, details on the training data of each of the four evaluated end-to-end ML docking tools are provided in the Supplementary Information (Lists S1–S4) to facilitate potential unbiased retraining and independent evaluation of these frameworks in future work.

2. Results and Discussion

2.1. Computational workflow

The computational workflow of our study is outlined in Fig. 1. First, a list of experimentally resolved protein-ligand complex structures was retrieved from the PDB using the “Advanced Search” option, combined with in-house filtering rules (May 2025). These rules primarily assessed the structural quality of the complexes as well as key parameters related to the drug-likeness of the co-crystallized small molecules (cf. “Experimental”). Selected PDB entries were prepared using custom Python and Bash pipelines, incorporating ChimeraX (v.1.8) and PyMOL (v.2.5), to ensure robust and fully automated large-scale curation. The resulting data set provides: (i) a comprehensive, machine-readable data table (csv) for downstream analysis, and (ii) thoroughly prepared protein (mol2) and ligand (mol2 and sdf) files, all chemically and structurally validated and directly ready for computational use. A new, up-to-date benchmark for docking power evaluation was then established as follows. The curated PDB entries were first assigned to either a training set or a test set using a strict cold-ligand split, which excluded any overlap of crystallographic ligands across partitions. This split considered both ligand HET codes and 2D molecular similarity, thereby eliminating potential data leakage. To facilitate model training not only on the full training set (which was relatively large) but also under progressively harder generalization scenarios, the training set was further divided into nine progressively smaller subsets. These subsets were constructed using a similarity-aware design, whereby training PDB entries containing ligands with higher maximum similarity to test-set ligands were progressively excluded, yielding subsets of decreasing size and increasing dissimilarity relative to the test set. Redocking of each co-crystallized ligand into its corresponding binding pocket was then performed (cf. “Experimental”). We employed both

Smina (v.2019-10-15), an enhanced fork of AutoDock Vina that relies on non-ML sampling algorithms,²⁸ and several SOTA end-to-end ML docking tools, namely DeepDock,¹⁵ Interformer,¹⁹ SurfDock,²⁰ and Uni-Mol Docking v.2.²¹ Importantly, no post-docking energy minimization or pose optimization was applied, in order to directly assess the intrinsic ability of each algorithm’s sampling procedure to generate near-native, physically and chemically plausible poses. Docking pose rescoring was conducted exclusively on Smina-generated redocked poses using GNINA 1.3, a CNN-based SF,¹³ across both training and test sets. Based on the Smina docking scores and GNINA 1.3 CNN scores of redocked poses from the training set and each of its nine subsets, we trained a series of 10 logistic regression classifiers. Each model was trained to discriminate near-native from non-native poses (defined by RMSD thresholds, cf. “Experimental”), using only the docking and CNN scores as input features, standardized with scikit-learn’s ‘StandardScaler’ (v.1.6.1).

Docking power was finally evaluated on the test set by comparing: (i) raw Smina docking scores, (ii) GNINA 1.3 CNN-based scores, (iii) the 10 logistic regression models trained on the training set and its subsets, and (iv) the four end-to-end ML docking tools. Evaluation criteria included RMSD of top-ranked poses relative to the crystallographic reference and PoseBusters compliance.²² All source code and input/output data, including the full benchmark (prepared structures, docking output, and derived features), are freely available in our GitHub repository and on Zenodo (cf. “Data Availability”).

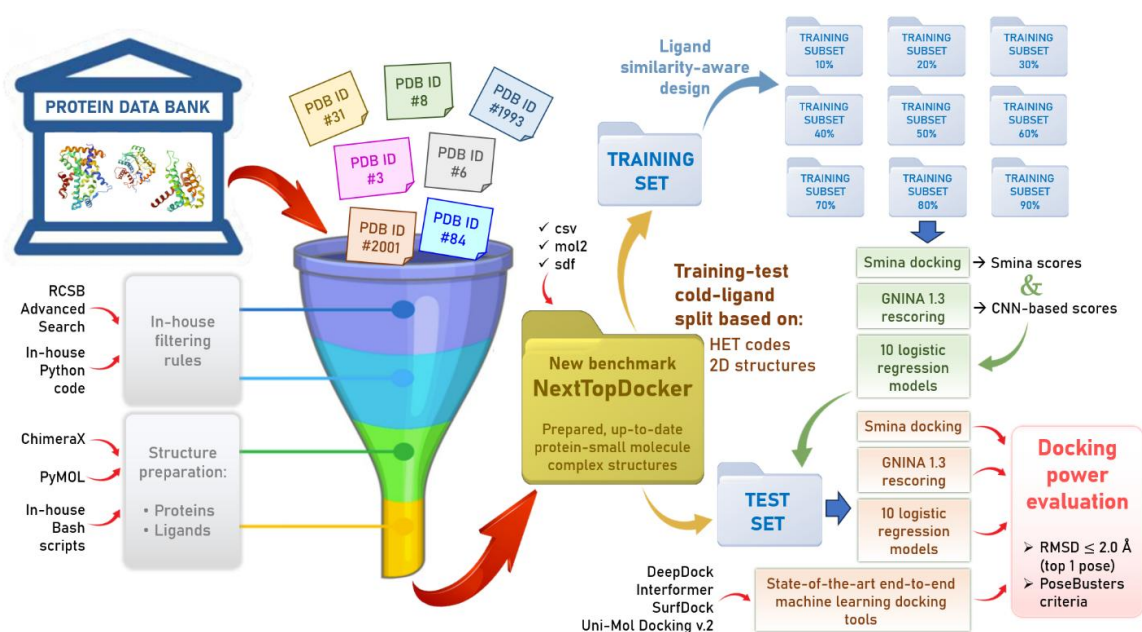


Fig. 1. Computational workflow of our study. An up-to-date, open-source benchmark for docking power evaluation, “NextTopDocker,” was built from experimentally solved protein-

ligand complexes automatically retrieved, curated, and prepared from the Protein Data Bank (May 2025), along with their Smina-generated redocked poses. A strict cold-ligand split, considering both ligand HET codes and 2D structures, was applied to define training and test sets. From the training set, nine progressively smaller, ligand-similarity-aware subsets were derived. Docking power was evaluated on the test set using Smina scores, GNINA 1.3 CNN scores, 10 logistic regression models trained on these scores across the training set and its subsets, and four SOTA end-to-end ML docking tools (DeepDock, Interformer, SurfDock, Uni-Mol Docking v.2). Evaluation criteria included RMSD of top-ranked poses against crystallographic ones and compliance with PoseBusters quality checks.

2.2. Data retrieval, data curation and stratified data split

Our final data set comprised 19,239 protein-ligand complexes retrieved, curated, and prepared from the PDB (cf. “Experimental”). It exhibited high diversity in terms of co-crystallized ligands and protein targets, as well as the functional families they represented.

In total, 13,750 unique ligands (distinct HET codes) were included. These ligands showed substantial structural diversity: among the 94,524,375 possible ligand pairs, only 1.85% had Tanimoto similarity coefficients (Morgan fingerprints, 2,048 bits, radius 2) greater than 0.2. The pairwise similarity distribution had a mean of 0.1105, further confirming the overall structural dissimilarity within the ligand set (Fig. S1, Supplementary Information). Moreover, these ligands spanned a broad chemical space, as reflected by the value ranges of 11 computed physicochemical properties (Fig. S2, Supplementary Information), while also satisfying an extended set of drug-likeness criteria (cf. “Experimental”).

On another note, 3,173 unique protein targets (distinct UniProt identifiers) were included and subsequently grouped into 261 functional families based on UniProt classification, an approach widely accepted for protein family assignment (Fig. S3, Supplementary Information). The most represented families were transferases (24.98%), hydrolases (21.10%), oxidoreductases (11.47%), and transcription factors (5.22%), each accounting for over 1,000 complexes. The most frequent targets were 3C-like proteinase (P0DTD1, 596 entries), carbonic anhydrase II (P00918, 331 entries), and bromodomain-containing protein 4 (O60885, 301 entries).

A strict cold-ligand split, considering both ligand HET codes and 2D molecular similarity constraints, was applied to divide the data set into training (14,038 PDB entries) and test (5,201 PDB entries) sets (cf. “Experimental”). The ligands across these two partitions were structurally dissimilar: among 36,683,616 ligand pairs that could be formed between the training ($n = 10,128$) and test ($n = 3,622$) ligands, only 1.71% had Tanimoto similarity scores above 0.2 (Fig.

2A). The highest Tanimoto similarity (Morgan fingerprints, 2,048 bits, radius 2) of a training ligand to any test ligand was centered around 0.3753 and did not exceed 0.99 (Fig. 2B), confirming minimal risk of data leakage or bias between the two sets. Note that this cold-ligand split was adopted because the logistic regression models and subsequent docking power evaluations would operate directly on ligand poses (cf. “Experimental”).

To further assess generalization under increasingly stringent conditions and to simulate realistic low-similarity scenarios, nine progressively dissimilar training subsets were derived from the full training set (cf. “Experimental”). As shown in Fig. S4 (Supplementary Information), nearly all ligands in the smallest training subset (10% of the training complexes) exhibited Tanimoto similarity coefficients below 0.2 to any ligand in the full test set. This design ensured that a model trained on such a small, chemically distant subset and still performing well on the full test set would demonstrate strong generalization and robust docking power.

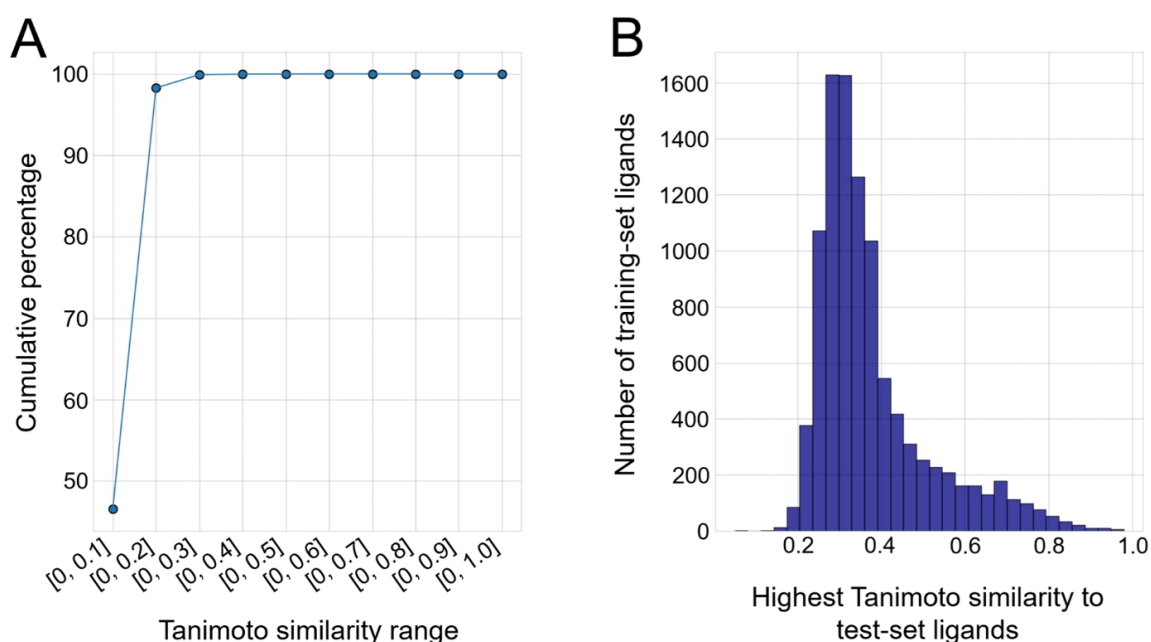


Fig. 2. Structural dissimilarity between training-set and test-set ligands. (A) Cumulative percentage of training-set and test-set ligand pairs whose Tanimoto similarity coefficients (Morgan fingerprints, 2,048 bits, radius 2) fell within defined ranges. (B) Distribution of the highest Tanimoto similarity coefficient of each training-set ligand to any ligand in the test set.

2.3. Docking power evaluation

2.3.1. Analysis of Smina-generated redocked poses

Redocking of all co-crystallized ligands into their native binding sites was performed using Smina (v.2019-10-15) under a rigid-receptor/flexible-ligand protocol,²⁸ based on preprocessed

protein and ligand 3D structures (cf. “Experimental”). For each protein target, the search space was defined by a series of cubic grid boxes centered on the centroid of the corresponding co-crystallized ligand, with increasing edge lengths of 10, 20, 30, and 40 Å (volumes of 1,000 to 64,000 Å³; up to 20 poses generated per size). This hierarchical design decoupled the docking search space from ligand size and experimental orientation, thereby avoiding the bias inherent to distance-based box definitions commonly used by end-to-end ML docking frameworks such as DeepDock and Interformer.^{15,19} The structural deviation between each redocked and crystallographic conformation was then quantified as the heavy-atom RMSD (Å), with poses classified as near-native (RMSD $\leq$ 2.0 Å) or non-native (RMSD $>$ 2.0 Å).

A total of 1,160,559 redocked poses were generated by Smina across 19,239 PDB entries. The number of poses generated per entry was strongly negatively correlated with the percentage of near-native poses ($r = -0.7089$, $p\text{-value} < 0.05$; Fig. 3A). This indicates that entries for which Smina produced a larger number of poses tended to yield a smaller fraction of near-native ones, suggesting that more extensive sampling often corresponds to more challenging docking cases, where high-quality binding geometries are harder to reproduce. Among the 68,565 near-native poses successfully generated across all entries, nearly 80% ($n = 53,879$) satisfied all PoseBusters validity criteria, including chemical, intramolecular, and intermolecular plausibility. A strong positive correlation was observed between the percentage of near-native poses and the percentage of poses that were both near-native and PoseBusters-valid per entry ($r = 0.8882$, $p\text{-value} < 0.05$; Fig. 3B), indicating that entries generating more near-native poses also tended to yield a higher proportion of physicochemically valid poses among them. Fig. 3C shows the density of near-native and PoseBusters-valid poses (red) and near-native but PoseBusters-invalid poses (blue), both exhibiting two peaks, indicative of bimodal distributions. Based on this observation, Fig. 3D presents the RMSD densities of PoseBusters-valid and PoseBusters-invalid near-native poses within the [0.0, 1.0] (Å) and (1.0, 2.0] (Å) intervals. In each RMSD range, the two distributions differed significantly (two-sided Mann-Whitney U test with Bonferroni correction, $p\text{-value} < 0.05$). Moreover, within each range, the median RMSD of PoseBusters-valid poses was lower than that of invalid ones, suggesting that physicochemically valid poses tended to be closer to the crystallographic binding mode. These findings reinforce that, although increasing the number of poses generated by Smina may also raise the number of non-native and PoseBusters-invalid poses, within the near-native population, poses that are more tightly redocked (lower RMSD) tend to exhibit higher physicochemical validity.

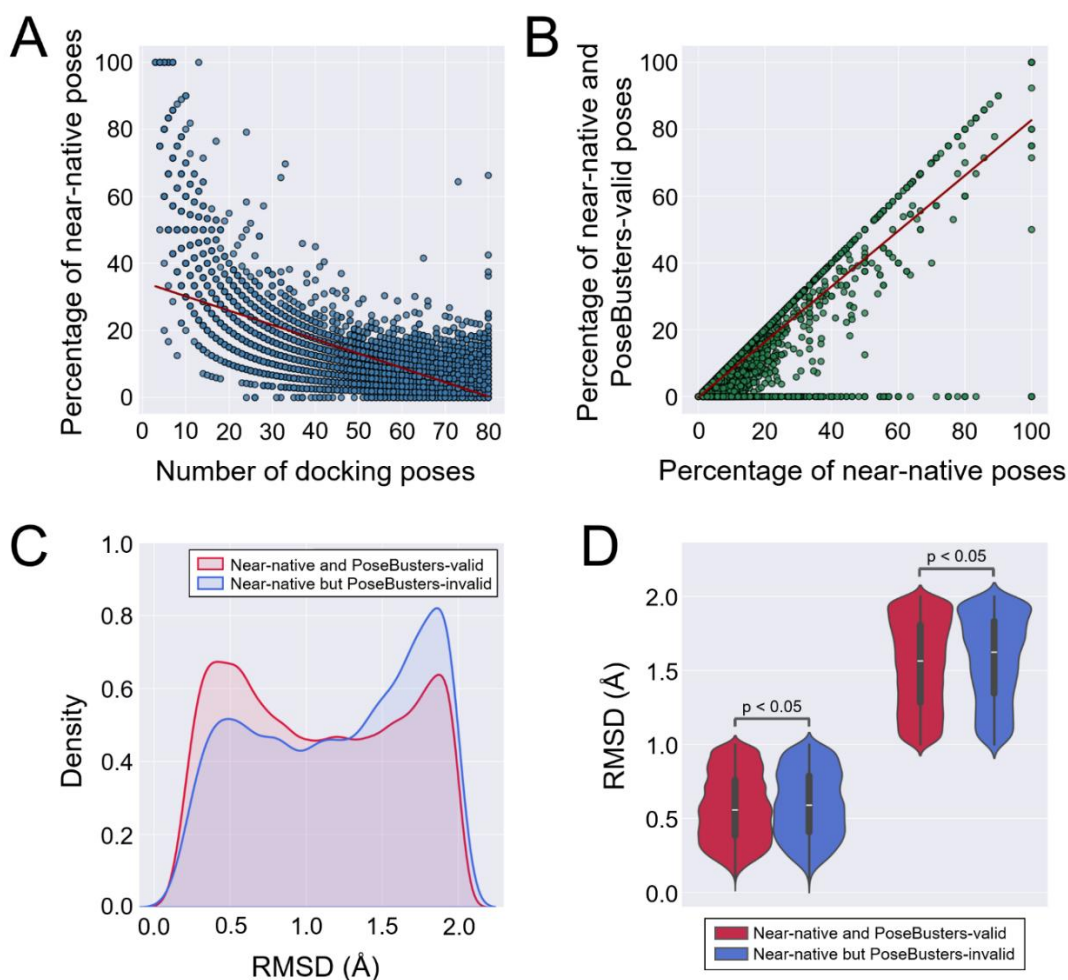


Fig. 3. Analysis of Smina-generated redocked poses. (A) Relationship between the number of poses generated per PDB entry and the percentage of near-native poses ($\text{RMSD} \leq 2.0 \text{ \AA}$), revealing a statistically significant negative correlation ($r = -0.7089$, $p\text{-value} < 0.05$). (B) Correlation between the percentage of near-native poses and the percentage of poses that were both near-native and PoseBusters-valid per PDB entry ($r = 0.8882$, $p\text{-value} < 0.05$). (C) RMSD density distributions of near-native and PoseBusters-valid poses (red) and near-native but PoseBusters-invalid poses (blue). (D) RMSD density distributions of PoseBusters-valid and PoseBusters-invalid near-native poses within the $[0.0, 1.0] \text{ (\AA)}$ and $(1.0, 2.0] \text{ (\AA)}$ intervals (two-sided Mann-Whitney U test with Bonferroni correction, $p\text{-value} < 0.05$), with PoseBusters-valid poses showing lower median RMSDs.

2.3.2. Docking power evaluation on the final benchmarking data set

The docking power of Smina, GNINA 1.3 (CNN-based rescoring), logistic regression rescoring on Smina-generated poses, and four end-to-end ML docking frameworks (DeepDock, Interformer, SurfDock, and Uni-Mol Docking v.2) was evaluated according to near-nativeness ($\text{RMSD} \leq 2.0 \text{ \AA}$) and PoseBusters physicochemical validity criteria on our full test set of 5,201

PDB entries (Table 1, Fig. 4). For Interformer, docking power was assessed using both Pose energy and Pose score metrics. Each of the 10 logistic regression models was trained on one of 10 progressively reduced, ligand-similarity-aware training splits, including the full training set. As input features, we used the scaled Smina docking score and the CNN-based score from GNINA 1.3 for each redocked pose in the corresponding training data (cf. “Experimental”). Importantly, GNINA 1.3 CNN-based rescoring and our logistic regression models were both applied exclusively to Smina-generated poses; hence, all conformations were sampled purely through physics-based docking, without any AI-driven pose generation. To ensure fair comparison across methods, all poses were evaluated directly, without any post-docking energy minimization or refinement, thereby reflecting the intrinsic docking power, i.e., the pose generation and ranking capability, of each docking framework. In line with the CASF-2016 definition of docking power,²⁶ only one top-ranked pose per test-set entry (according to each method’s internal scoring) was retained for evaluation, preventing artificial inflation of performance due to inclusion of multiple top poses.

Table 1. Docking power of Smina, GNINA 1.3 (CNN-based rescoring) and logistic regression rescoring on Smina-generated poses, as well as four end-to-end ML docking frameworks (DeepDock, Interformer, SurfDock, and Uni-Mol Docking v.2), evaluated on our full test set.

Docking framework/model [†]	Median RMSD of top-ranked poses across entries (Å) [#]	Percentage of entries with a near-native pose ranked first [#]	Percentage of entries with a near-native and PoseBusters-valid pose ranked first [#]
Smina	1.1922	62.74	53.47
GNINA 1.3	1.0441	71.01	59.68
LogReg (100%)	0.9935	73.60	61.93
LogReg (90%)	0.9942	73.52	61.91
LogReg (80%)	0.9942	73.52	61.91
LogReg (70%)	0.9942	73.54	61.91
LogReg (60%)	0.9935	73.60	61.93
LogReg (50%)	0.9913	73.60	61.93
LogReg (40%)	0.9913	73.60	61.95
LogReg (30%)	0.9913	73.60	61.95
LogReg (20%)	0.9852	73.76	62.14
LogReg (10%)	0.9831	73.81	62.24
DeepDock	5.5899	7.42	1.50
Interformer (Pose energy)	0.9951	75.24	63.78
Interformer (Pose score)	0.9635	77.49	64.78
SurfDock	0.8944	83.08	43.07
Uni-Mol Docking v.2	1.3171	72.39	6.33

[†]LogReg (x%): Logistic regression model trained on the x% training (sub)set.

[#]Only one top-ranked pose per test-set entry (according to each method’s internal scoring) was retained for evaluation.

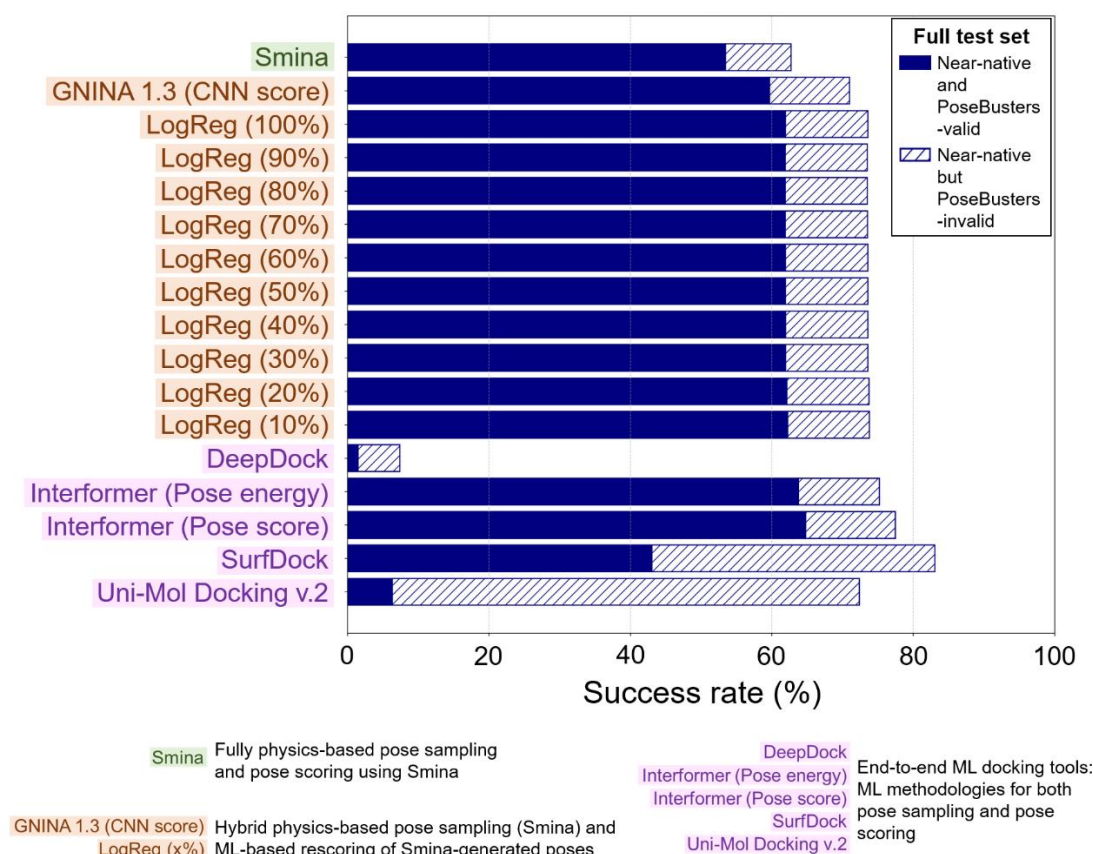


Fig. 4. Docking power of Smina, GNINA 1.3 (CNN-based rescoring) and logistic regression rescoring on Smina-generated poses, as well as four end-to-end ML docking frameworks (DeepDock, Interformer, SurfDock, and Uni-Mol Docking v.2), expressed as success rate (%) according to near-nativeness ($\text{RMSD} \leq 2.0 \text{ \AA}$) and PoseBusters physicochemical validity checks, evaluated on our full test set (5,201 PDB entries). The docking power of Interformer was assessed using both Pose energy and Pose score metrics. Each of the 10 logistic regression models was trained on one of 10 progressively reduced, ligand-similarity-aware training splits, including the full training set, and is denoted as “LogReg (x%),” where x% indicates the proportion of the original training data used for model training.

When physicochemical plausibility was taken into account, three of the four end-to-end ML docking tools (DeepDock, SurfDock, Uni-Mol Docking v.2) failed to produce valid poses for more than half of the test set. Among these, only Interformer achieved performance comparable to our logistic regression-based rescoring models, regardless of whether Pose energy or Pose score was used for pose selection. However, when considering only top-1 poses per test-set entry that were both near-native and PoseBusters-valid, all 10 of our logistic regression models recovered significantly more key non-covalent interactions (hydrogen bonds, ionic contacts, π - π stacking), analyzed using the ProLIF Python library (v.2.0.3),²⁹ than Interformer (Fig. 5, two-

sided Mann-Whitney U tests with Bonferroni correction, p -values < 0.01). For each such pose generated by Smina and selected by our logistic regression models, an average of 80% of key interactions were reproduced relative to the corresponding crystal ligand, compared with 72–74% for Interformer-generated poses. Moreover, nearly 60% of top-1 poses prioritized by each logistic regression model achieved a 100% interaction recovery rate relative to the ground-truth pose, versus only 45–47% for Interformer. These results indicate that our simple rescoring models more reliably prioritize poses that are not only geometrically accurate and physicochemically plausible, but also faithfully reproduce the key protein-ligand interactions observed in experimental structures.

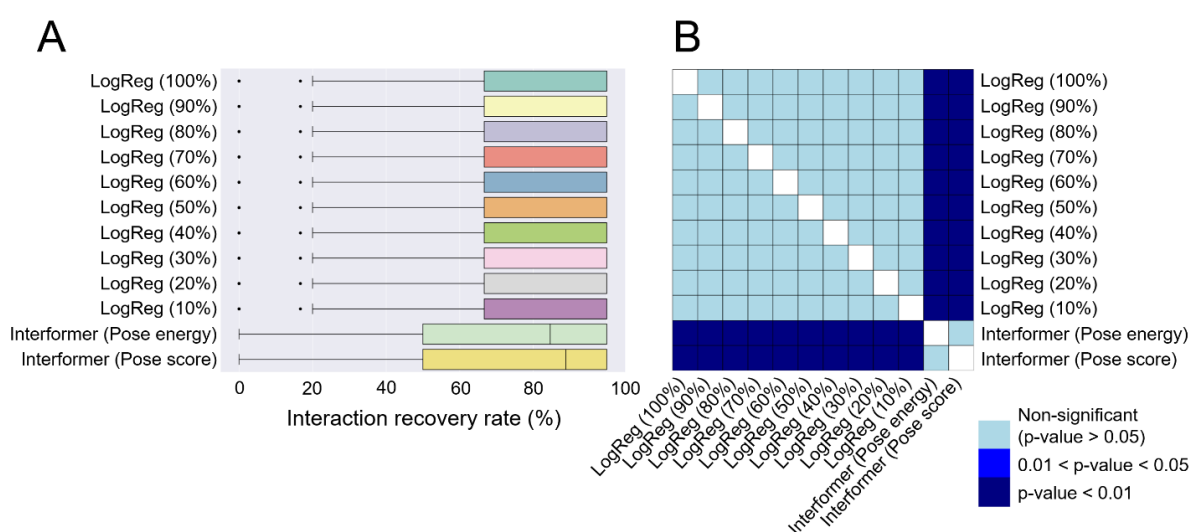


Fig. 5. Ability of our 10 logistic regression models and Interformer (Pose energy and Pose score metrics) to recover key non-covalent interactions (hydrogen bonds, ionic contacts, π - π stacking) among top-1 poses per test-set entry that were both near-native and PoseBusters-valid. (A) Boxplots showing the distribution of interaction recovery rates (%) across top-1 near-native and PoseBusters-valid poses. Our hybrid Smina pose sampling-ML pose scoring workflow achieved an average recovery rate of 80%, compared with 72–74% for Interformer (Pose energy and Pose score). (B) Heatmap showing pairwise p -values obtained from two-sided Mann-Whitney U tests with Bonferroni correction, comparing interaction recovery rates between models. The rates achieved by our 10 logistic regression models did not differ significantly from each other, demonstrating the robustness of our workflow despite reductions in training data. In contrast, each of our models differed significantly from Interformer (p -values < 0.01). Together with the average recovery rates, these results indicate that our hybrid workflow consistently recovered significantly more key non-covalent interactions relative to ground-truth poses than Interformer.

On another note, DeepDock exhibited particularly poor results even under the RMSD-only criterion, while SurfDock and Uni-Mol Docking v.2 displayed a clear dependency on physics-based, post-docking minimization: without this refinement step, SurfDock often generated poses suffering from severe steric clashes with protein atoms, and Uni-Mol Docking v.2 frequently produced distorted ligand geometries with incorrect bond lengths and angles (Fig. S5, Supplementary Information). Similar observations were made when the aforementioned docking approaches were evaluated on their respective cold test subsets, generated from the full test set by excluding PDB entries containing ligands with a Tanimoto similarity ≥ 0.99 (Morgan fingerprints, 2,048 bits, radius 2) to any ligand included in that tool's training data (Figs. S6 and S7, Supplementary Information). The reliance of SurfDock on post-docking minimization was, in fact, explicitly demonstrated in its original publication (cf. Fig. 2 of the SurfDock paper),²⁰ where the tool alone performed worse than several classical docking programs on the PoseBusters and Astex benchmarks. Similarly, Uni-Mol Docking v.2's dependence on a "post-processing" refinement functionality has even been raised in its GitHub repository (e.g., <https://github.com/deepmodeling/Uni-Mol/issues/358>, accessed October 2025, among other issues).

Our 10 logistic regression models were trained solely using near-native and non-native labels defined by RMSD, without any information from PoseBusters physicochemical criteria, ensuring a fair comparison with end-to-end ML docking approaches. Since each model outputs a probability score from a binary classifier, these models can be regarded as classification-based pose scorers that achieved balanced accuracies exceeding 0.85 in both 5-fold cross-validation (CV) and the full test set (Table S1, Supplementary Information). This demonstrates that such simple models, trained only on Smina and GNINA 1.3 scores, are capable not only of distinguishing near-native from non-native poses but also of ranking PoseBusters-valid conformations at the top with performance comparable to, or much better than, SOTA DL-based pose generation methods. Remarkably, even when trained on smaller, ligand-similarity-aware subsets, our logistic regression models retained full docking power on the complete test set, both in terms of near-nativeness combined with PoseBusters validity (Table 1, Fig. 4) and interaction recovery rate (Fig. 5), underscoring the robustness of this straightforward rescoring approach and the strong complementarity between physics-based sampling and lightweight ML-based pose scoring.

Moreover, we examined how robust our conclusions were to changes in the RMSD threshold defining near-native poses. For this purpose, we evaluated two additional thresholds: 1.0 Å and 0.5 Å. When a 1.0-Å threshold was applied, the same trend as with 2.0 Å was observed: among

the four end-to-end ML docking tools, only Interformer matched the performance of our 10 logistic regression models, yielding comparable proportions of test-set entries for which a redocked pose with $\text{RMSD} \leq 1.0 \text{ \AA}$ and satisfying all PoseBusters criteria was ranked first (Fig. S8A, Supplementary Information). Interestingly, with a stricter $0.5\text{-\AA}$ threshold, our hybrid Smina docking-ML scoring workflow ranked the largest fraction of such ultra-near-native poses as top-1 (nearly 25% without PoseBusters checks and still over 20% when these were enforced). In contrast, Interformer and SurfDock ranked such highly accurate and PoseBusters-valid poses first for only about 15% of test-set entries, whereas DeepDock and Uni-Mol Docking v.2 fell below 5% (Fig. S8B, Supplementary Information). This once again highlights the superior robustness and generalization of hybrid physics-ML docking-scoring strategies over current end-to-end generative architectures.

2.4. Discussion

This work presents a rigorous and up-to-date evaluation of molecular docking performance based on a large-scale, meticulously curated benchmarking data set comprising 5,201 prepared PDB entries in the test set and 14,038 entries in the training set, each with Smina-generated docking poses, hereafter referred to as “NextTopDocker.” To our knowledge, this is the first time that such an extensive, chemically diverse and structurally representative benchmark, updated as of May 2025, has been employed to assess the docking power of four SOTA end-to-end ML docking frameworks: DeepDock (2021),¹⁵ Interformer (2024),¹⁹ SurfDock (2025),²⁰ and Uni-Mol Docking v.2 (2025).²¹ The inclusion of this broad and contemporary data set ensures that the evaluation reflects the current state of structural biology, rather than relying on outdated or limited benchmarks that may not capture the growing complexity of ligand-protein interactions found in the PDB. Beyond its scale and diversity, the relevance of this benchmark also lies in reinforcing a key principle of SB drug design: the ability to recover experimentally observed binding poses remains a cornerstone of rational ligand optimization. Accurate reproduction of the ground-truth binding mode ensures that key protein-ligand interactions, often involving residues critical for biological activity, are correctly captured. In contrast, erroneous poses may occupy irrelevant binding sites or form non-physiological contacts, limiting interpretability and potentially misleading subsequent optimization efforts. Reliable pose prediction therefore provides the structural foundation upon which fragment-based and structure-guided hit-to-lead campaigns are built, underscoring the continued importance of benchmarking docking power as a core performance metric for both physics-based and ML-driven docking workflows.

Even though our logistic regression models were trained on subsets containing only a few thousand PDB entries, with ligands exhibiting low similarity to those in the test set, a small proportion of near-native poses, and no influence from PoseBusters filtering criteria, they achieved docking power surpassing that of most of the aforementioned SOTA ML docking tools. This held true whether performance was assessed by RMSD alone or by RMSD combined with PoseBusters physical plausibility checks. Furthermore, they recovered significantly more key non-covalent interactions among top-1 near-native and PoseBusters-valid poses than Interformer, the only end-to-end ML docking tool that matched their docking power. These results highlight the strong capability of physics-based, non-ML sampling algorithms in efficiently exploring conformational space, as well as the synergy between physics-based SFs (Smina) and ML-derived SFs (GNINA 1.3 CNN scores) for accurate pose ranking. Importantly, our use of a cold-ligand split and subsequent similarity-aware subdivision of the training data did not adversely affect test-set docking power, demonstrating the robustness of our simple yet interpretable approach. The success of such lightweight models underscores the importance of physical realism and robust ML design over architectural complexity in molecular docking.

One key methodological distinction of our benchmark lies in its search space definition, which ensures an unbiased and realistic evaluation of docking accuracy. Several ML-based docking frameworks such as Interformer and DeepDock define their search space as an expansion around the crystallographic ligand by a fixed distance (e.g., 10 Å from the ligand's heavy atoms).^{15,19} Similarly, Uni-Mol Docking v.2 defines a cubic grid based on the ligand's molecular size with a uniform 10-Å margin.²¹ While these strategies simplify the sampling process, they also artificially favor recovery of the ground-truth pose, as the model is guided by prior knowledge of the crystal ligand's position and orientation. This setup effectively reduces the problem's difficulty and can lead to inflated estimates of docking power. In contrast, our workflow adopted fixed grid box sizes, independent of the ligand's size or experimental pose orientation, thus ensuring a more rigorous and unbiased redocking assessment that better reflects real-world docking scenarios. Note that, in our study, two of the four fixed grid box sizes, with side lengths of 30 Å and 40 Å, already exceeded the cubic search space used in the PoseBusters study, which was defined as a cube (25 Å per side) centered on the geometric centroid of the crystal ligand's heavy atoms.²² Both our setup and that of PoseBusters positioned the grid box center at this same centroid, ensuring consistency in spatial alignment. However, by adopting larger grid dimensions in several configurations, our workflow provided a broader conformational search space, even more unbiased and rigorous than the PoseBusters setup, which its authors already described as "comparable with blind docking."²² This design further

reinforces that the docking accuracy observed in our evaluation reflects genuine sampling-scoring performance, rather than being aided by artificially constrained search spaces.

A notable weakness shared among many end-to-end ML docking tools is their dependence on post-generation refinement steps. Frameworks such as SurfDock and Uni-Mol Docking v.2 require physics-based energy minimization or pose optimization after pose generation to achieve acceptable performance; otherwise, their docking accuracy drops markedly.^{20,21} This reliance indicates that their intrinsic pose sampling mechanisms remain suboptimal, unable to consistently produce physically plausible poses without external correction. Although modern diffusion- and generative-based docking architectures have demonstrated remarkable ability to sample diverse conformations and explore large configurational spaces, they still lack explicit physical constraints governing molecular interactions during the generative process.²² Consequently, the poses generated often exhibit steric clashes, unrealistic bond geometries, or violations of hydrogen-bonding patterns, as previously highlighted by the PoseBusters study.²² This stems from a fundamental limitation of current data-driven paradigms: they optimize for statistical fidelity to training data distributions, not for energetic or physicochemical realism.^{23,30–33} Hence, while these models may capture global structural trends, they frequently overlook the underlying energetic landscape that determines molecular stability and binding feasibility. Our findings thus reinforce the need for physics-informed constraints to be incorporated directly within ML-based pose generation frameworks,^{33–35} rather than relying on post-docking refinements that merely compensate for inadequacies in the sampling process.

Among the evaluated ML docking methods, DeepDock warrants particular discussion. DeepDock was trained and evaluated using a time-based split between training and test data, a strategy now known to introduce training-test bias.²² Temporal splits commonly used in training-test partitioning can introduce data leakage or contamination, leading to overestimation of model performance.^{36,37} Structurally similar complexes resolved within close time frames can share homologous proteins or closely related ligands, meaning that a model may learn correlations based on these coincidental similarities rather than genuine predictive patterns.³⁷ Moreover, database updates are uneven across targets and chemotypes, meaning that chronological separation only loosely reflects the true progression of structural and chemical novelty. In such cases, performance metrics evaluated on temporally split data sets can overestimate a model's true generalization ability, giving a biased impression of its docking power or pose selection accuracy.³⁷ Accordingly, DeepDock's high performance reported on its original benchmark cannot be reliably extrapolated to unseen data that do not carry this temporal bias. Numerous subsequent independent studies have, indeed, shown that when

evaluated on other test sets, DeepDock’s performance declines sharply, often falling below that of even classical, purely physics-based docking algorithms such as Gold, Vina and Smina.^{19–22,38–40} Our findings corroborate this observation, demonstrating that ML models trained under biased conditions may generalize poorly to chemically and structurally dissimilar systems. Interestingly, the fact that DeepDock and other end-to-end ML docking tools can perform worse than non-ML, physics-based docking workflows underscores an important point: there remains substantial room for improvement in ML-based pose sampling. Despite recent progress, these methods still struggle to reconcile physical validity with generative efficiency.²² Their strong reliance on implicit priors and data-driven distributions, without explicit energy-based reasoning, limits their capacity to produce chemically meaningful poses across diverse targets and ligand classes.²³ In contrast, our results reaffirm that classical docking algorithms, when combined with robust, interpretable ML frameworks for pose scoring and selection, can deliver reliable and computationally efficient solutions to the docking problem. Overall, this work offers both a comprehensive benchmark for advancing future ML and hybrid docking methods and a conceptual framework that aligns with recent efforts to critically reassess the direction of the field.^{22,37,41,42} Comparative performance trends across purely physics-based, ML-based, and hybrid methodologies indicate that the primary bottleneck in molecular docking lies not in pose generation, but rather in pose scoring and selection.^{2,10} Achieving more accurate and transferable docking predictions will therefore require the integration of physically grounded principles. One promising avenue is the development of hybrid physics-informed neural architectures. For example, a diffusion model could be pre-trained on PoseBusters-valid docking poses generated by classical, physics-based sampling methods, such as the Smina-generated poses in the training set of our “NextTopDocker” benchmark, and subsequently fine-tuned and evaluated on our test set. This approach would enable the model to capture both the energetic realism and the statistical diversity inherent to biomolecular recognition.

3. Conclusions

In this study, we introduce the largest-scale, up-to-date (as of May 2025), and fully open-access benchmark for docking power assessment. “NextTopDocker,” our curated data set of 19,239 PDB-derived protein-ligand complexes, split into 14,038 training and 5,201 test entries via a strict cold-ligand strategy, together with nine ligand-similarity-aware training subsets, provides a challenging, diverse, and reproducible foundation for evaluating pose generation and docking

performance. This benchmark fills a critical gap left by widely used but limited data sets such as Astex, CASF, and PoseBusters, which, due to their small size (only hundreds of PDB entries), do not adequately capture the chemical and structural diversity of modern docking problems. Our comprehensive evaluation of four SOTA end-to-end ML docking frameworks: DeepDock (2021), Interformer (2024), SurfDock (2025), and Uni-Mol Docking v.2 (2025) revealed systematic weaknesses:

- (i) Their training-test partitions are often ambiguous or biased, sometimes based on time-dependent splits that permit training-test leakage;
- (ii) The definition of their search spaces, especially for pocket-oriented methods such as DeepDock, Interformer or Uni-Mol Docking v.2, relies on the co-crystal ligand’s 3D size and orientation, which artificially increases the likelihood of near-native pose generation;
- (iii) The resulting poses are frequently physically or chemically implausible, as echoed by PoseBusters analyses; and
- (iv) Several frameworks (e.g., SurfDock and Uni-Mol Docking v.2) rely on post-generation physics-based (non-ML) energy minimization or pose optimization to achieve acceptable accuracy, thereby masking shortcomings in their intrinsic ML-based sampling capability.

In contrast, our simple logistic regression models, trained on Smina and GNINA 1.3 scores from chemically dissimilar ligands and applied to Smina-generated poses, achieved docking power comparable to or exceeding that of the four SOTA end-to-end ML docking tools. Even when trained on minimal and low-similarity data, these models generalized robustly across over 5,000 test complexes, underscoring the resilience of physics-based sampling and hybrid scoring. Altogether, our findings highlight the limitations of current ML-based pose generation and emphasize the importance of embedding physics-informed constraints directly within ML docking architectures to ensure physically valid poses without reliance on post-docking refinement. Furthermore, our results suggest that future progress in molecular docking will depend more critically on improving pose scoring and selection than on pose generation alone.

4. Experimental

Data retrieval and curation

A raw data table containing information on experimental structures was first downloaded from the PDB,⁴³ using the “Advanced Search” option with the following criteria: (i) the structure must be compatible with the pdb format; (ii) the polymer entity type must be “Protein” and the entry polymer type must be “Protein (only);” (iii) the refinement resolution must be ≤ 2.5 Å;

(iv) the experimental method must be “X-ray diffraction;” (v) the refinement R factors, “R-value free” and “R-value work,” must be lower than 0.30 and 0.25, respectively;⁴⁴ and (vi) the UniProt accession code must not be empty. As of May 2025, a total of 126,961 PDB entries satisfied these selection criteria.

The data table obtained from the above search was subsequently processed as follows:

(i) PDB entries containing unknown ligands (“UNL”) or lacking co-crystallized ligands (apo-structures) were removed;

(ii) Only entries with an R-factor discrepancy (R-value free – R-value work) between 0 and 0.05 were retained;⁴⁴

(iii) Inorganic ligands were removed, defined as those containing at least one atom other than C, H, O, N, P, S, F, Cl, Br, or I;

(iv) PDB entry ID-ligand ID pairs were retained only if the ligand satisfied an extended set of drug-likeness criteria:

- Molecular weight (MW) between 250 and 800 Da (values taken directly from the PDB);
- Lipophilicity (WlogP) between –2.0 and +6.0 (calculated using the ‘rdkit.Chem.Crippen’ module implemented in RDKit, v.2025.03.3);
- Number of rotatable bonds lower than 15 and topological polar surface area smaller than 180 Å² (calculated using ‘rdkit.Chem.rdMolDescriptors’);
- Numbers of hydrogen bond acceptors and hydrogen bond donors were each lower than 10 (calculated using ‘rdkit.Chem.Lipinski’).

These physicochemical ranges were chosen to capture bioactive, non-fragment molecules while covering a broader ligand space than conventional filters (e.g., Lipinski’s Rule of Five or Ghose’s filter). The criteria included modern drugs such as kinase inhibitors, antivirals, macrocycles, natural products, and allosteric modulators, while maintaining tractability for downstream docking with Smina;^{45–47}

(v) Solvents, buffers, additives, detergents, and coenzymes were removed;

(vi) Complexes containing invalid ligands due to missing atoms were excluded;

(vii) In complexes that contained more than one ligand after Steps (i)–(vi), the ligand exhibiting a therapeutically relevant phenotype on the target was chosen as the redocked ligand.

Each remaining PDB complex was then prepared according to the following procedure:

(i) Protein file preparation:

An in-house Bash script was developed to automate the following tasks:

- From each original PDB entry, the protein structure (with associated metal ions, if present in the binding site) was extracted and saved in pdb format;

- To better mimic a realistic binding environment, metal ions were retained if located within 5 Å of any heavy atom of the co-crystallized ligand, provided they were supported by Smina's built-in atom type definitions and/or considered biologically relevant (e.g., involved in catalysis or ligand coordination, rather than crystallization artifacts). Retained ions included: Ca, Cd, Co, Cu, Fe, K, Mg, Mn, Mo, Na, Ni, and Zn;
- Target structures were then prepared using ChimeraX's 'DockPrep' tool (v.1.8). Hydrogen atoms and charges were added according to the atomic type of each hydrogen-bearing heavy atom. Non-complex ions (those not covalently or coordinatively bound in the binding site) were deleted. Incomplete side chains were rebuilt using the Dunbrack rotamer library. Final protein structures were saved in mol2 format.

(ii) Ligand file preparation:

A PyMOL (v.2.5) batch script was created to automate the following tasks:

- Each complex structure was fetched by its PDB ID, and the co-crystallized ligand was extracted using its HET code and chain identifier;
- Hydrogens were added in PyMOL, and the ligand was saved in both mol2 and sdf formats;
- Ligand sanitization was checked using the 'SanitizeMol' function from RDKit's 'Chem' module (v.2025.03.3). If sanitization failed, the ligand's experimental 3D conformation (sdf format) was downloaded manually from the RCSB database, hydrogens were re-added in PyMOL, and the molecule was saved again in mol2 and sdf.

Each PDB entry successfully prepared and retained in our data table corresponds to a protein target co-crystallized with a drug-like ligand of therapeutic relevance for that protein. For each entry, the data table includes: (i) the PDB ID; (ii) the co-crystallized ligand's HET code, MW (Da), IUPAC name, SMILES string, WLogP, number of rotatable bonds, numbers of hydrogen bond donors and acceptors, and topological polar surface area (Å²); (iii) the R-value free, R-value work, and R-factor discrepancy; (iv) the entry resolution (Å); and (v) keywords describing the protein family of the target and the phenotype of the co-crystallized ligand. This data table is provided as a csv file, along with all prepared protein and ligand structures (proteins: mol2; ligands: mol2 and sdf) on GitHub and Zenodo (cf. "Data Availability").

Stratified data split

Since our approach would be evaluated on its docking power, i.e., its ability to operate on the 3D conformations of ligands generated by docking, the training-test data split (at the level of PDB entries) followed a strict cold-ligand principle. Specifically, no co-crystallized ligand HET code was shared between the training and test sets. However, because distinct HET codes do

not necessarily preclude high structural similarity (e.g., enantiomers with identical 2D fingerprints), we further enforced a 2D cold-ligand constraint. In the training set, only PDB entries whose ligands had a maximum Tanimoto similarity (Morgan fingerprints, 2,048 bits, radius 2) ≤ 0.99 to any ligand in the test set were retained.^{6,48} This additional filtering ensured that no training-test bias or ligand-level data leakage occurred. To further challenge the models, we adopted a similarity-aware design by partitioning the training set into nine subsets with sizes ranging from 90% to 10% of the full training set, in 10% decrements. Subsets of decreasing size were created by progressively excluding PDB entries whose co-crystallized ligands exhibited Tanimoto similarity (Morgan fingerprints, 2,048 bits, radius 2) above successively lower thresholds relative to the test-set ligands (Table 2). Thus, the smaller the subset, the more structurally dissimilar its ligands were to those in the test set. This design allowed us to evaluate model performance not only as a function of training set size but also under increasing distributional shift, mimicking realistic scenarios in which ligands for novel targets differ substantially from those in the training data. Note that, since multiple training-set ligands could share identical maximum similarity scores to test-set ligands, exact 10% splits could have required random selection among tied training-set PDB entries. To avoid introducing such randomness, we instead allowed subset sizes to deviate slightly from the target percentages (e.g., 9.63%, 20.03%, 30.08%), which were subsequently rounded to the nearest 10% for reporting purposes. On another note, tool-specific cold test subsets were constructed by removing from the full test set all entries whose ligands exhibited a Tanimoto similarity (Morgan fingerprints, 2,048 bits, radius 2) ≥ 0.99 to any ligand included in the training data of each end-to-end ML docking tool. This procedure yielded four test subsets: DeepDock-cold, Interformer-cold, SurfDock-cold, and Uni-Mol Docking v.2-cold, ensuring unbiased evaluation by preventing any data overlap between training and testing (Table 2).

Table 2. Composition of the training set, its nine similarity-aware subsets, the test set and its four tool-specific cold subsets.

Data partition	Number of unique PDB entries	Number of co-crystallized ligands (unique HET codes)	Highest similarity between each training-(sub)set ligand and all test-set ligands [#]
<i>Training set</i>			
100%	14,038	10,128	0.9839
90% (89.96%)	12,628	9,517	0.6620
80% (80.08%)	11,241	8,794	0.5349
70% (70.00%)	9,826	7,828	0.4382
60% (60.03%)	8,427	6,757	0.3793
50% (50.00%)	7,019	5,592	0.3431

40% (40.10%)	5,629	4,443	0.3175
30% (30.08%)	4,222	3,266	0.2963
20% (20.03%)	2,812	2,051	0.2747
10% (9.63%)	1,352	822	0.2481
<i>Test set</i>			
100%	5,201	3,622	
DeepDock-cold	4,284	3,451	
Interformer-cold	4,175	3,420	Not applicable
SurfDock-cold	4,248	3,442	
Uni-Mol Docking v.2-cold	3,899	3,366	

[#]Tanimoto similarity according to Morgan fingerprints (2,048 bits, radius 2)

Molecular docking with Smina and rescoring using GNINA 1.3

We performed redocking of all co-crystallized ligands into their respective binding sites using a rigid-receptor/flexible-ligand protocol with Smina (v.2019-10-15).²⁸ Docking was carried out on the 3D mol2 structures of proteins and ligands previously prepared (cf. “Data retrieval and curation”). For each protein target, the binding pocket was defined by a grid box centered at the centroid of the corresponding co-crystal ligand (‘autobox_ligand’ option). The ‘exhaustiveness’ parameter was set at 8 (default value), and a fixed number of seeds was appointed, ensuring that our docking results would be reproducible. To increase the diversity of generated poses for subsequent analyses, we tested four grid box volumes: 1,000 Å³ (10 Å × 10 Å × 10 Å), 8,000 Å³ (20 Å × 20 Å × 20 Å), 27,000 Å³ (30 Å × 30 Å × 30 Å), and 64,000 Å³ (40 Å × 40 Å × 40 Å). This strategy ensured that the search space was not dependent on the size or the experimentally solved orientation of the co-crystallized ligand. In doing so, we avoided the bias introduced by the more conventional search-space definition, where a fixed distance is added around each heavy atom of the crystallographic conformation. The latter approach inherently favors recovery of the ground-truth pose, thereby making the redocking process artificially easier, whereas our setup provided a more challenging and unbiased evaluation of docking accuracy. For each grid box size, up to 20 poses were retained in both mol2 and sdf formats, resulting in a theoretical maximum of 80 poses per ligand. The ‘minimize’ option was not used during docking to avoid additional post-docking minimization of poses. The RMSD (Å) between each redocked conformation and the crystallographic pose (heavy atoms only) was computed using Open Babel’s ‘obrms’ tool (v.3.1.1).⁴⁹ Redocked conformations with RMSD ≤ 2.0 Å from the experimental pose were classified as near-native, while those with RMSD > 2.0 Å were considered non-native.

Since the number of non-native redocked poses largely exceeded that of near-native ones, we alleviated this imbalance in the training set by reducing redundancy among non-native poses.

Specifically, two non-native poses (from the same PDB entry) were considered redundant if their pairwise RMSD did not exceed 0.5 Å. In such cases, the pose with the higher RMSD relative to the experimental conformation was discarded. This filtering step was applied exclusively to the training set. The test set, on the other hand, was left unmodified, as all generated poses must be retained to ensure a fair comparison between our docking workflow and end-to-end ML docking methods. All remaining redocked poses (mol2 format) in the training set, as well as all poses in the test set, were rescored using GNINA 1.3 with the ‘score_only’ option.¹³ The protein conformations were identical to those used in previous Smina docking runs, while the ligand poses were retained exactly as generated by Smina. Default parameters were used for CNN-based scoring, and a “CNNScore” was recorded for each redocked pose. All docking poses (mol2, sdf), along with the corresponding Smina docking scores and GNINA 1.3 CNN-based scores are provided in our GitHub repository and on Zenodo (cf. “Data Availability”).

Evaluation of docking poses using PoseBusters

We assessed the physical and chemical validity of redocked poses using the PoseBusters test suite, following the authors’ recommendations.²² Three input files were prepared for each complex: (i) an sdf containing the redocked ligand pose, (ii) an sdf of the corresponding crystal ligand (pre-checked for sanitization as described above), and (iii) a pdb file of the protein with only the co-crystallized ligand removed. A redocked pose was considered PoseBusters-valid if its RMSD to the crystal ligand did not exceed 2.0 Å and it passed all 26 PoseBusters criteria, grouped into three categories: (i) chemical validity and consistency (file loading, sanitization, InChI conversion, atom connectivity, molecular formula, molecular bonds, double bond stereochemistry, and tetrahedral chirality); (ii) intramolecular validity (bond lengths and angles, internal steric clashes, aromatic ring planarity, non-aromatic ring non-planarity, double bond planarity, and energy ratio); and (iii) intermolecular validity (overlapping volume and minimum distance between ligand atoms and protein atoms, organic or inorganic cofactors, and water molecules, as well as the maximum protein-ligand atom distance). Note that we computed RMSD values using the ‘obrms’ function in Open Babel (v.3.1.1),⁴⁹ ensuring consistency across all docking tools in this study.

Construction of logistic regression models

We used the ‘LogisticRegression’ function from scikit-learn (v.1.6.1) with default hyperparameters (including L2 regularization with $C = 1.0$) to train 10 logistic regression

models, each corresponding to the entire training set or one of its nine similarity-aware subsets. These models were trained to discriminate between near-native and non-native redocked poses, as defined by their RMSD. Using default hyperparameters is standard practice for logistic regression when interpretability and reproducibility are prioritized, especially since this model type has few sensitive hyperparameters and is not prone to overfitting in its regularized default form.⁵⁰

For each model, 5-fold CV was conducted on the corresponding training data. Each fold was defined using a cold-ligand split (based on the HET codes of the co-crystallized ligands) to prevent information leakage. For every redocked pose, its Smina docking score and GNINA 1.3 CNN-based score were normalized using the ‘StandardScaler’ function (default parameters) from scikit-learn (v.1.6.1): the scaler was fit on each training fold and applied to the corresponding validation fold. Since logistic regression is a binary classifier, the decision threshold was optimized to balance performance between classes. Out-of-fold predictions (i.e., the predicted probabilities of each pose being near-native) were collected across the five CV folds, and the optimal threshold was defined as the probability cutoff that maximized the mean balanced accuracy across folds. This optimization was performed using Optuna (v.4.4.0) with a search space ranging from 0.01 to 0.99, aiming to maximize the mean balanced accuracy. The Tree-structured Parzen Estimator (TPE) sampler was used with a fixed random seed to ensure reproducibility. The number of optimization trials was set to 100, as recommended in the Optuna documentation.⁵¹

After CV, each logistic regression model was retrained on its entire training data and evaluated on the test set to assess generalizability. The ‘StandardScaler’ was fit on the corresponding complete training data (either the full training set or one of its subsets) and subsequently applied to the test set. The decision threshold determined during CV was then used to compute the confusion matrix and balanced accuracy. For each PDB entry, the Smina redocked pose with the highest predicted probability of being near-native was retained for docking power evaluation.

DeepDock

Input files for DeepDock consisted of the protein structures (in pdb format) and their respective co-crystallized ligands (in sdf), both prepared following the same protocol used for the PoseBusters benchmark.²² Using these two files, a polygon mesh representing the molecular surface of the binding site was generated with MSMS and PyMesh,⁵² employing the default parameters provided in the DeepDock GitHub repository (<https://github.com/OptiMaL-PSE->

[Lab/DeepDock](#), accessed August 2025). As described by the authors of DeepDock, the binding site was defined as all protein atoms within 10 Å of any ligand atom. The resulting mesh, saved in ply format, together with the crystal ligand, was used for redocking with the pre-trained DeepDock model and its corresponding checkpoint. For each PDB entry, DeepDock generated a single optimized conformation as an RDKit Mol object, which was subsequently saved in sdf using the ‘SDWriter’ function from the ‘rdkit.Chem’ module. In cases where the optimization process exceeded the maximum number of iterations but still produced a conformation that could be saved as an sdf, the pose was retained for evaluation of near-nativeness and physicochemical plausibility. For conformations that were successfully generated by DeepDock but could not be written by the ‘SDWriter’ function integrated into the DeepDock code (as provided by its authors), provided that the corresponding crystal ligand was successfully loaded and sanitized with RDKit, we considered them as dockable cases that subsequently failed to pass any criteria. The RMSD between generated and crystal poses was computed using Open Babel (v.3.1.1), and all resulting redocked poses were further evaluated with the PoseBusters test suite to assess their physicochemical validity.

Interformer

Input for Interformer consisted of the original protein-ligand complexes downloaded from the Protein Data Bank (May 2025, in pdb format) and the corresponding co-crystallized ligands in sdf.¹⁹ All input structures were prepared following the Interformer authors’ guidelines to ensure its optimal performance and to avoid any bias that could favor our own workflow in the comparison of docking power. For the complexes, only heteroatoms annotated as “UNX” were removed. Ligand structures in sdf were the same as those obtained from our previously described preparation procedure (cf. “Data retrieval and curation”). In cases where ligands could not be processed by Interformer, the crystallographic conformations were retrieved directly from the RCSB database and hydrogens were added using PyMOL (v.2.5). All 5,201 co-crystallized test-set ligands were successfully loaded with RDKit and passed its sanitization. As recommended in the Interformer literature, the co-crystallized conformation of each ligand was used as the starting structure, rather than a geometry relaxed by the universal force field (UFF). For each entry, 20 conformations were generated following the energy functions and checkpoints provided in the Interformer GitHub repository (<https://github.com/tencent-ailab/Interformer>, accessed August 2025). These conformations were used without modification as input to the pose-scoring model. To evaluate docking performance, for each entry, we selected the single conformation with either the lowest pose energy or the highest

pose score. RMSD values between the selected conformations and their corresponding experimental crystal structures were computed with Open Babel (v.3.1.1). Finally, to assess the fraction of successful cases that also satisfied structural validity criteria, generated protein (in pdb, without waters or co-crystallized ligands) and ligand files (in sdf) were subjected to PoseBusters analysis.

SurfDock

For protein structures, SurfDock employed a tri-level representation comprising the sequence level, the residue graph level, and the protein surface level. The sequence and residue graph embeddings were obtained using the large language model ESM-2,⁵³ while the surface representation was prepared following a protocol similar to that used in DeepDock and MaSIF.⁵⁴ As described by the authors of SurfDock, the binding site was defined as all protein atoms within 8 Å of any ligand atom. Before surface triangulation with MSMS and processing with PyMesh, each protein structure was prepared with Open Babel (v.3.1.1), all receptor hydrogens were corrected using Reduce,⁵⁵ and the resulting structure was saved in pdb format. For ligand structures, the co-crystallized conformations (in sdf) were prepared in the same manner as those used for the PoseBusters test suite. For each PDB entry in the test set, SurfDock generated 40 candidate conformations using the confidence score functions and checkpoints provided in the SurfDock GitHub repository (<https://github.com/CAODH/SurfDock>, accessed August 2025). To specifically assess the intrinsic pose sampling capability of the model, no post-docking energy minimization was performed: the ‘force_optimize’ argument was disabled, while all other parameters were kept at their default values. For docking power evaluation, the top-ranked conformation based on the highest confidence score was selected per test-set entry (in sdf). The RMSD between the selected pose and the corresponding crystal ligand was computed using Open Babel (v.3.1.1), after which the pose was further analyzed with the PoseBusters test suite to assess its physicochemical validity.

Uni-Mol Docking v.2

Input files for Uni-Mol Docking v.2 consisted of the template protein structures (in pdb format) and their corresponding co-crystallized ligands (in sdf). The preparation of the initial protein and ligand files followed the protocol described in the PoseBusters paper, which was also used for generating the PoseBusters benchmark.²² Specifically, from each test-set complex, only the co-crystallized ligand of interest (the one selected for redocking) was removed. This resulted in a protein structure that was identical to the original crystal complex except for the absence of

that single ligand, while all other structural components were retained. Ligands were extracted from their test-set complexes using PyMOL (v.2.5) and saved in sdf. All 5,201 were successfully loaded with RDKit and passed its sanitization prior to docking with Uni-Mol Docking v.2.

The docking grid box was defined based on the coordinates of the crystallographic ligand. The grid center corresponded to the centroid of all heavy atoms of the ligand, and the box dimensions were computed as the orthogonal extent of the ligand plus an additional 10-Å margin along each Cartesian axis. Consequently, the final grid box size depended on both the size and 3D conformation of the crystallographic ligand. Grid box information was stored in a json file. To evaluate the intrinsic pose sampling capability of Uni-Mol Docking v.2, no post-generation pose refinement (e.g., steric clash correction) was applied. Specifically, the ‘steric-clash-fix’ argument was disabled, and all other parameters were retained at their default settings, using the publicly available implementation of Uni-Mol Docking v.2 (https://github.com/deepmodeling/Uni-Mol/tree/main/unimol_docking_v2, accessed October 2025). For each test-set complex, a single docking pose was generated (in sdf). The RMSD between the predicted and crystallographic poses was computed using Open Babel (v.3.1.1). Finally, the docked ligand structures were evaluated with PoseBusters, alongside their corresponding protein templates, to assess the chemical and physical plausibility of the generated docking poses.

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